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MTH213/MAT213: Differential Equation

First Order Ordinary Differential Equation: Integrating Factor Method

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Linear First Order Equations

A first order equation is linear if it can be written in this form

$$y' + p(x)y = g(x) \tag{1.1}$$

Where both $p(x)$ and $g(x)$ are continuous functions.

Examples:

1. $y' = 3y + 3$
2. $x^2y' + 3xy = x^2$
3. $(x^2 + 1)y' + 3x^3y = 6e^{-\frac{3}{2}x^2}$

Example 2 is not in the standard form of (1.1), but is linear, since it can be rewritten as $y' + \dfrac{3}{x^2}y = 1$

Integrating Factor Solution Method

To solve a first order linear ordinary differential equation use the following steps:

- Put the differential equation in the standard form
- Identify $p(x)$
- Find the integrating factor using $\mu(x) = e^{\int p(x)dx}$
- Multiply both sides of the differential equation by $\mu(x)$
- Write the left hand side in the short form as $(\mu(x)y(x))'$
- Integrate both sides, make sure you add the constant of integration.
- Solve for the solution $y(x)$.

Linear First Order Equations Contd

Examples:

1. Find the general solution of $y' + 2y = x^3 e^{-2x}$

Solution: Integrating factor is $e^{\int 2dx} = e^{2x}$

Next multiply the equation by the integrating factor to obtain
$e^{2x}(y' + 2y) = e^{2x}(x^3 e^{-2x})$

$\Rightarrow (e^{2x}y)' = x^3 \Rightarrow \int (e^{2x}y)'dx = \int x^3 dx$ so that

$$e^{2x}y = \frac{x^4}{4} + c \Rightarrow y = \frac{x^4 e^{-2x}}{4} + ce^{-2x}$$

2. Find the solution of the initial value problem
$y' - 2y = te^{-2t}, \quad y(1)=0$

Linear First Order Equations Contd

Solution: The integrating factor is $e^{\int -2dt} = e^{-2t}$

$\Rightarrow e^{-2t}(y' - 2y) = e^{-2t}(te^{-2t}) \Rightarrow (ye^{-2t})' = te^{-4t}$

Next we simplify

$ye^{-2t} = \int te^{-4t}dt = -e^{-4t}\left(\frac{t}{4} + \frac{1}{16}\right) + c$. Therefore

$y = -e^{-2t}\left(\frac{t}{4} + \frac{1}{16}\right) + ce^{2t}$.

To find c we use the given initial condition

$0 = -e^{-2}\left(\frac{1}{4} + \frac{1}{16}\right) + ce^{2} \Rightarrow c = \frac{5e^{-4}}{16}$. So that

$y = -e^{-2t}\left(\frac{t}{4} + \frac{1}{16}\right) + \frac{5e^{-4}e^{2t}}{16}$

Linear First Order Equations Contd

Problems:

Solve the following

1. $y' + \frac{1}{x}y = e^{x^2}$

2. $(x^2+1)\frac{dy}{dx} + 3xy = 6x$

3. $\cos x\, y' + \sin x\, y = 2\cos^3 x \sin x - 1 \quad y\left(\frac{\pi}{4}\right) = 3\sqrt{2}, \quad 0 \le x < \frac{\pi}{2}$

4. $ty' - 2y = t^5 \sin 2t - t^3 + 4t^4 \quad y(\pi) = \frac{3}{2}\pi^4$

5. $xy' + 5y = 7x^2, \quad y(2) = 5$

6. $y' = (1-y)\cos x, \quad y(\pi) = 2$

Linear First Order Equations Contd

7. $y' + 2y = te^{-2t}, \quad y(1) = 0$
8. $2xy' - 3y = 9x^3$
9. $(x+1)y' - y = e^x(x+1)^2$
10. $(x^3 - x)y' - (3x^2 - 1)y = x^5 - 2x^3 + x$